MAVEN TOYS SALES ANALYSIS

End-to-end data pipeline project

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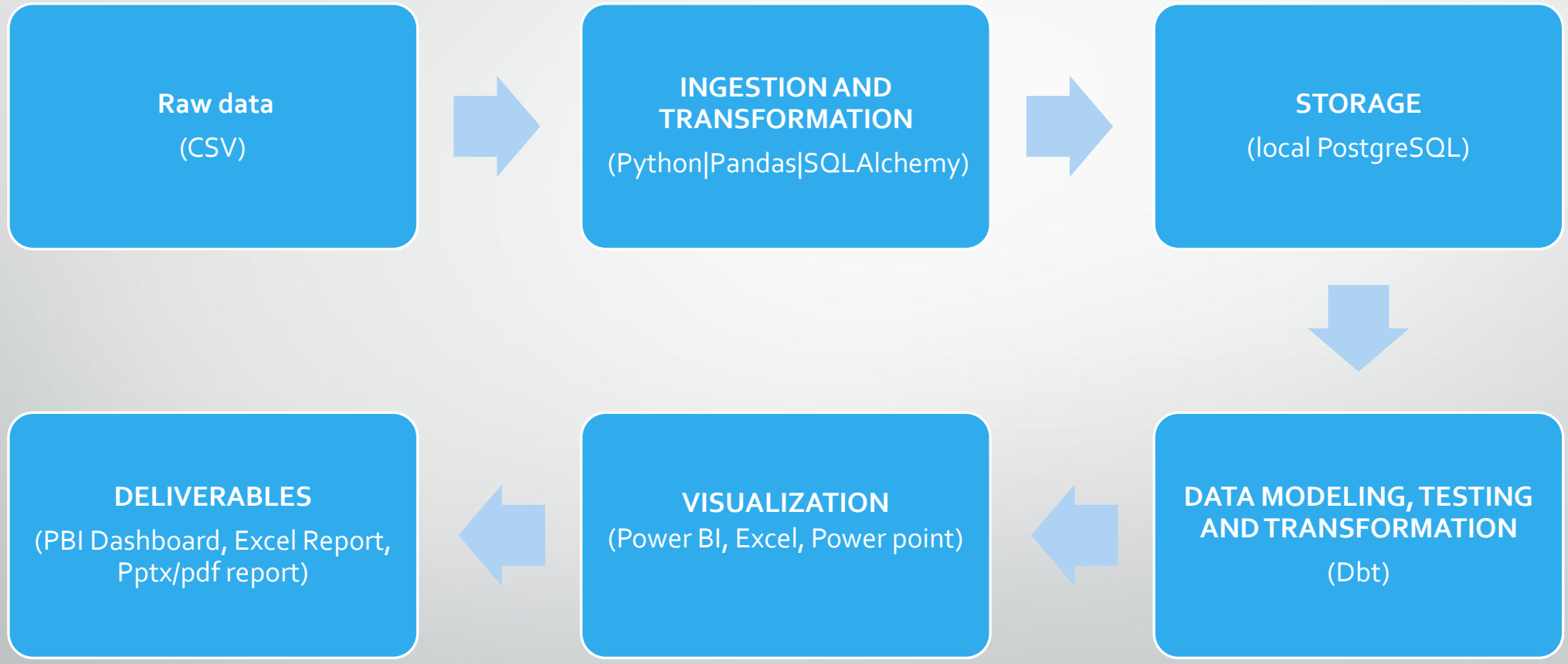
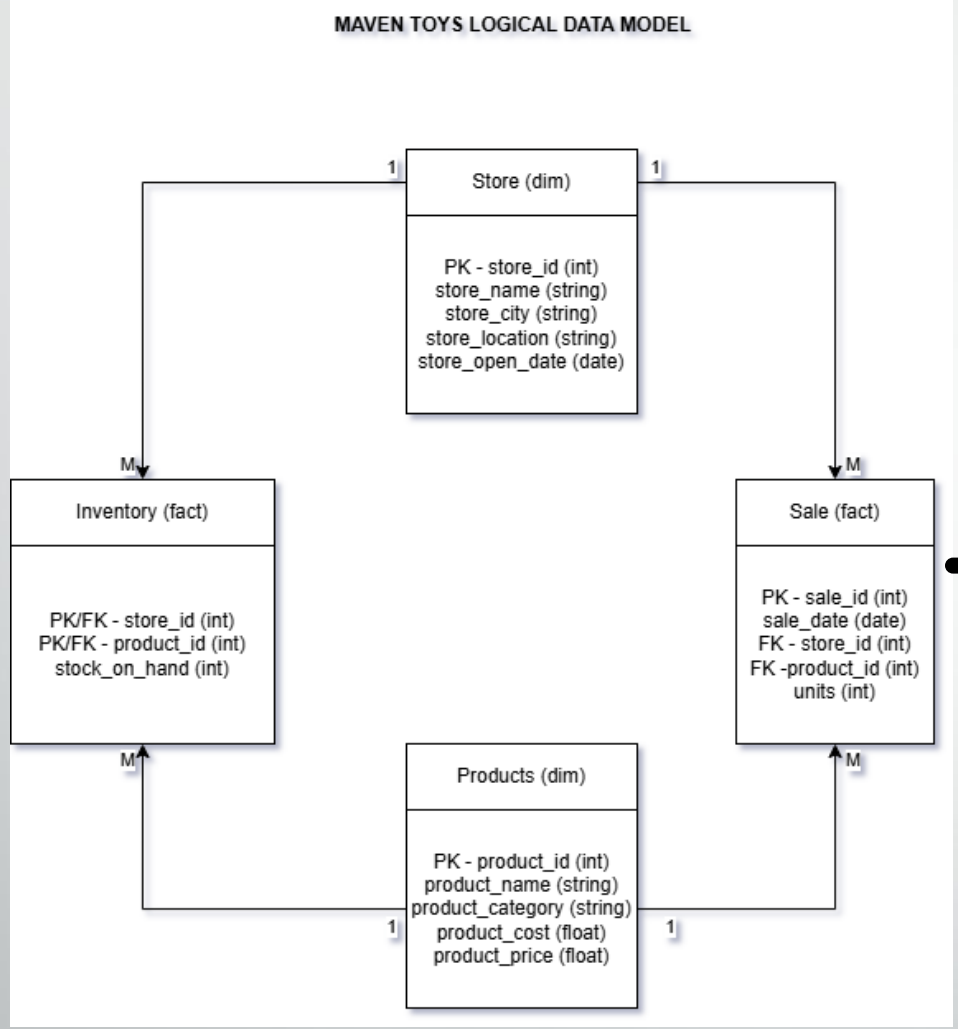
PROJECT OVERVIEW

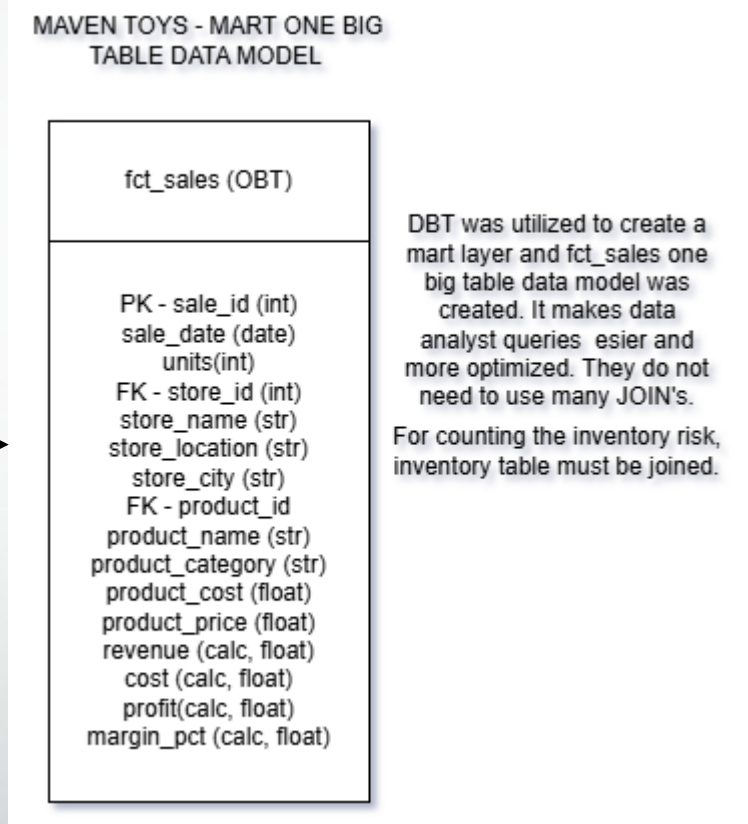
- **Business model:** Maven toys is a toys retail company in Mexico, which is operating stores in different cities and locations and it sells different toy products.
- **The data:** 5 tables. The sales table has around 829k rows and stores past sale data from January 2022 to September 2023.
- **THE MAIN GOAL: To understand what business decisions they could derive from their (past) data, which might convert to more revenue and profit in the future.**
- **Secondary goal: To create a structure for data automatization process, so that the results could be delivered more efficiently and more sustainably.**

DATA PIPELINE AND RESULT DELIVERING PROCESS



DATA MODEL – From star schema to one big table





A Star schema was ingested, which was big and inefficient. Layered with Dbt and created an OBT model with automatic metric counting. 4 more OBT models where created in marts layer.

OTHER TECHNICAL NUANCES

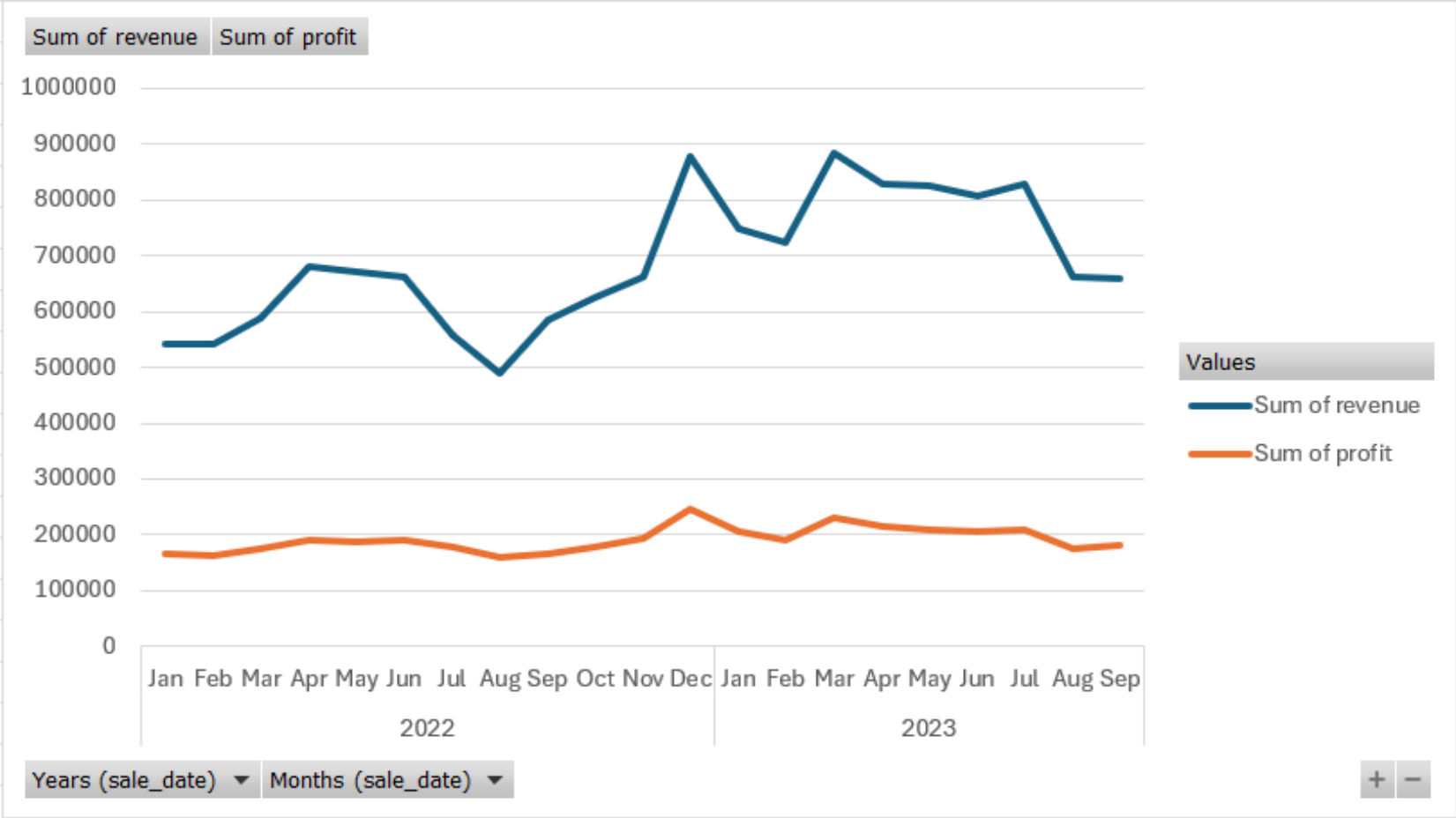
For replication of production environment Docker was implemented. 3 different containers where used.

Airflow was used for pipeline orchestration.

Dbt was used for model testing and dveloping a semantic layer.

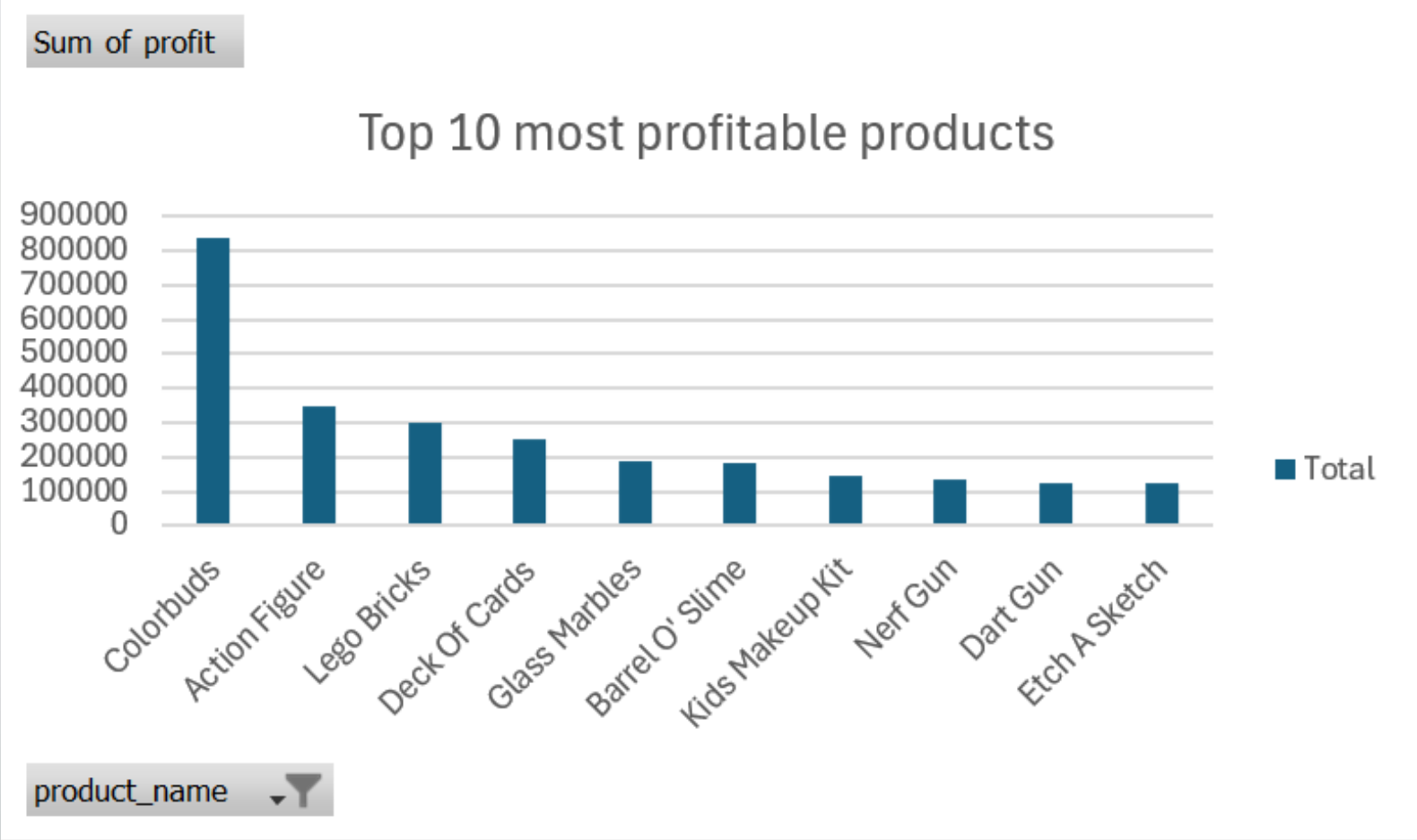
KEY FINDINGS

KF 1– REVENUE TRENDS



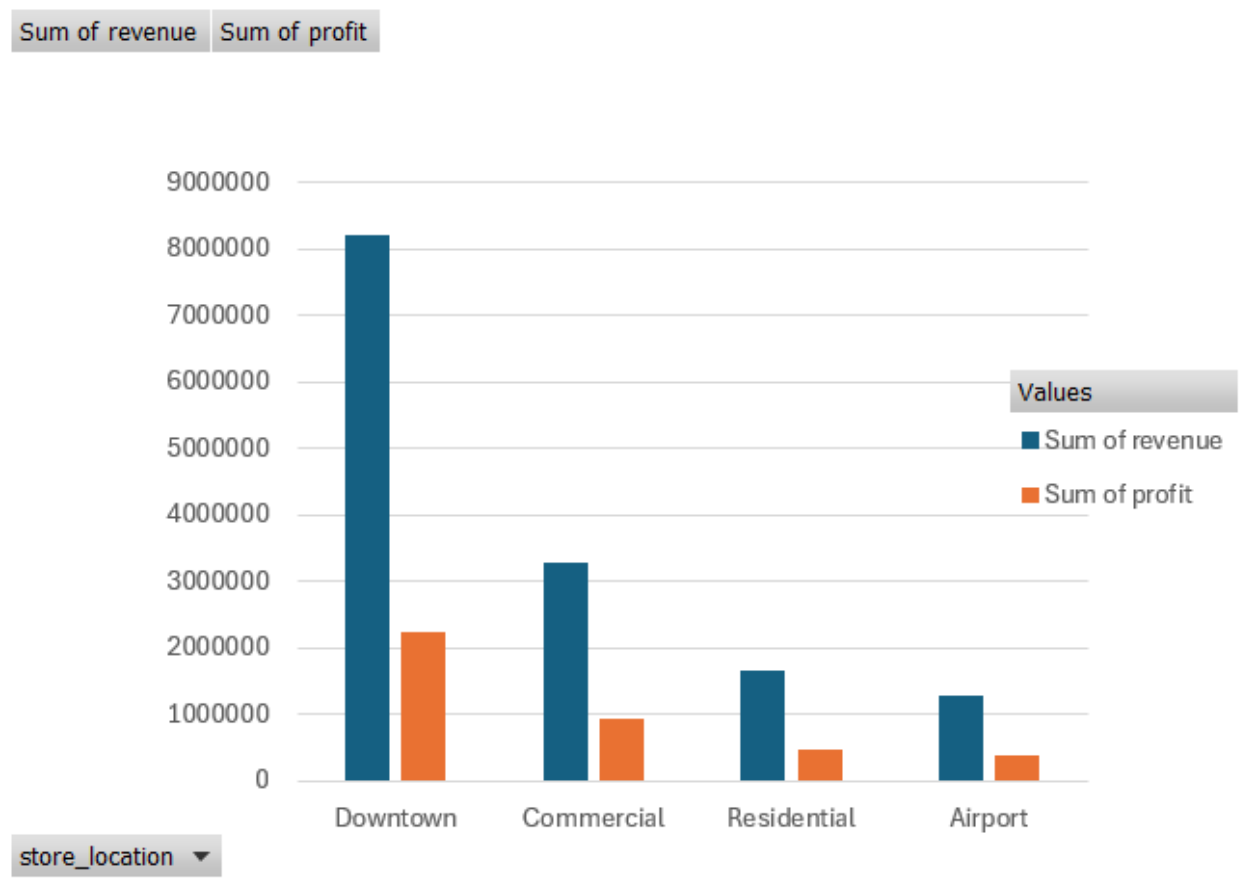
- A spike of revenue and profit during December can be seen. This is due to Christmas holiday season. In April its Día del Niño (Children‘s day) in Mexico.
- 2023 consistently stronger than 2022 - growth trend confirmed.
- Profit margin stable at ~25% despite revenue fluctuations.

KF 2 – PRODUCT PERFORMANCE



- Colorbuds (electronics) are making the most profit during the period. (Cost 6.99, Retail Price 14.99)
- Other products might need more marketing or discounts. Revenue generation must be diversified to lower the supply chain risks.
- Limitation: No marketing performance data.

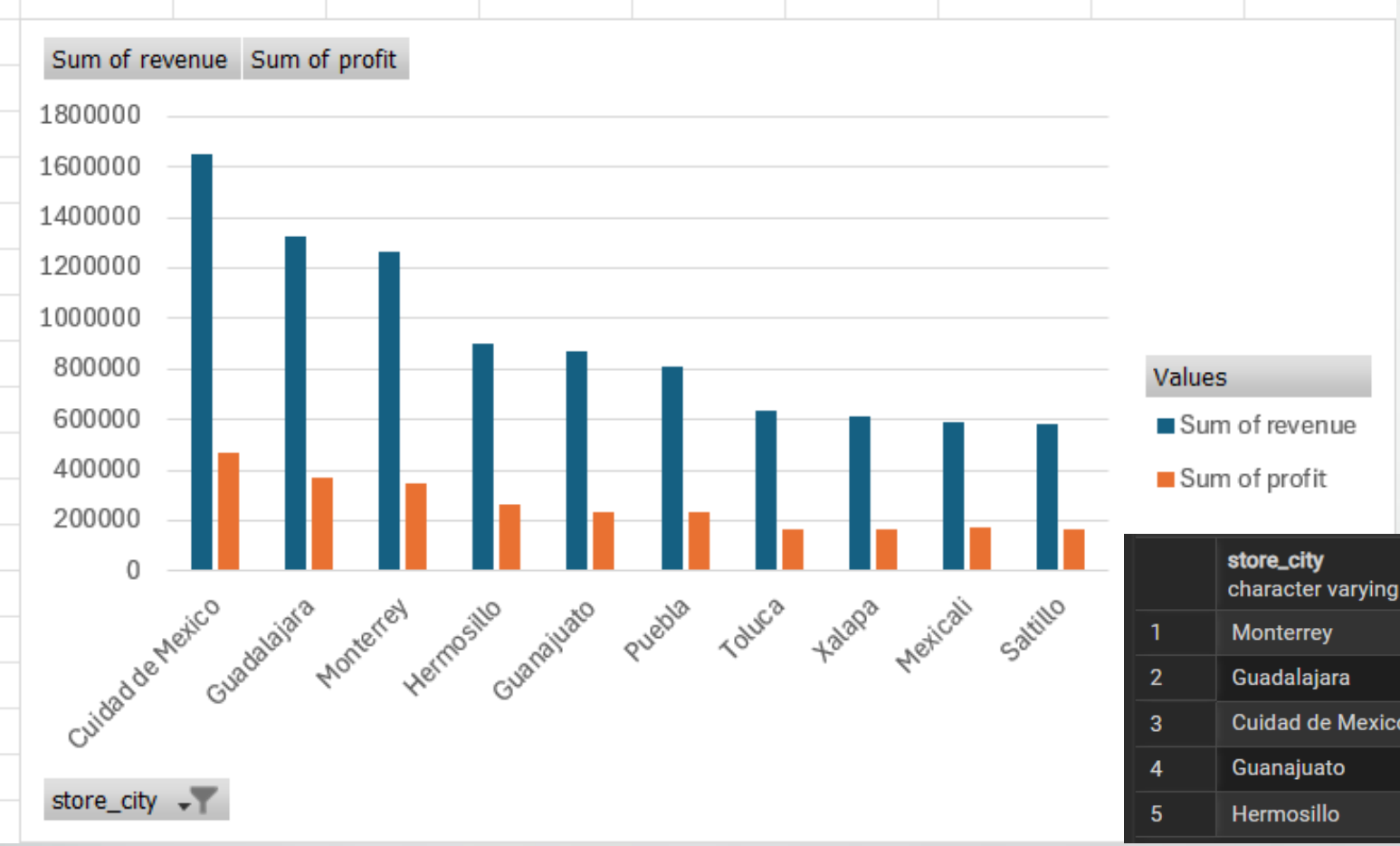
KF 3 – STORE LOCATION INSIGHTS



	store_location character varying (100)	store_count bigint	revenue_per_store numeric
1	Airport	3	429907.53
2	Residential	6	276019.00
3	Commercial	12	273261.63
4	Downtown	29	283434.35

- Downtown location generates the most revenue and profit.
- In Downtown 29 stores in Airport 3 stores.
- Airport location has a better performance on average per store.
- Limitation: No store operating costs data.

KF 4 – PERFORMANCE BY CITY



	store_city character varying (100)	store_count bigint
1	Monterrey	4
2	Guadalajara	4
3	Cuidad de Mexico	4
4	Guanajuato	3
5	Hermosillo	3

- Ciudad de Mexico (4 stores) the capital city generates the most revenue and profit. More population means more profit.
- More stores in the city means more revenue from the city. Altough population matters too.

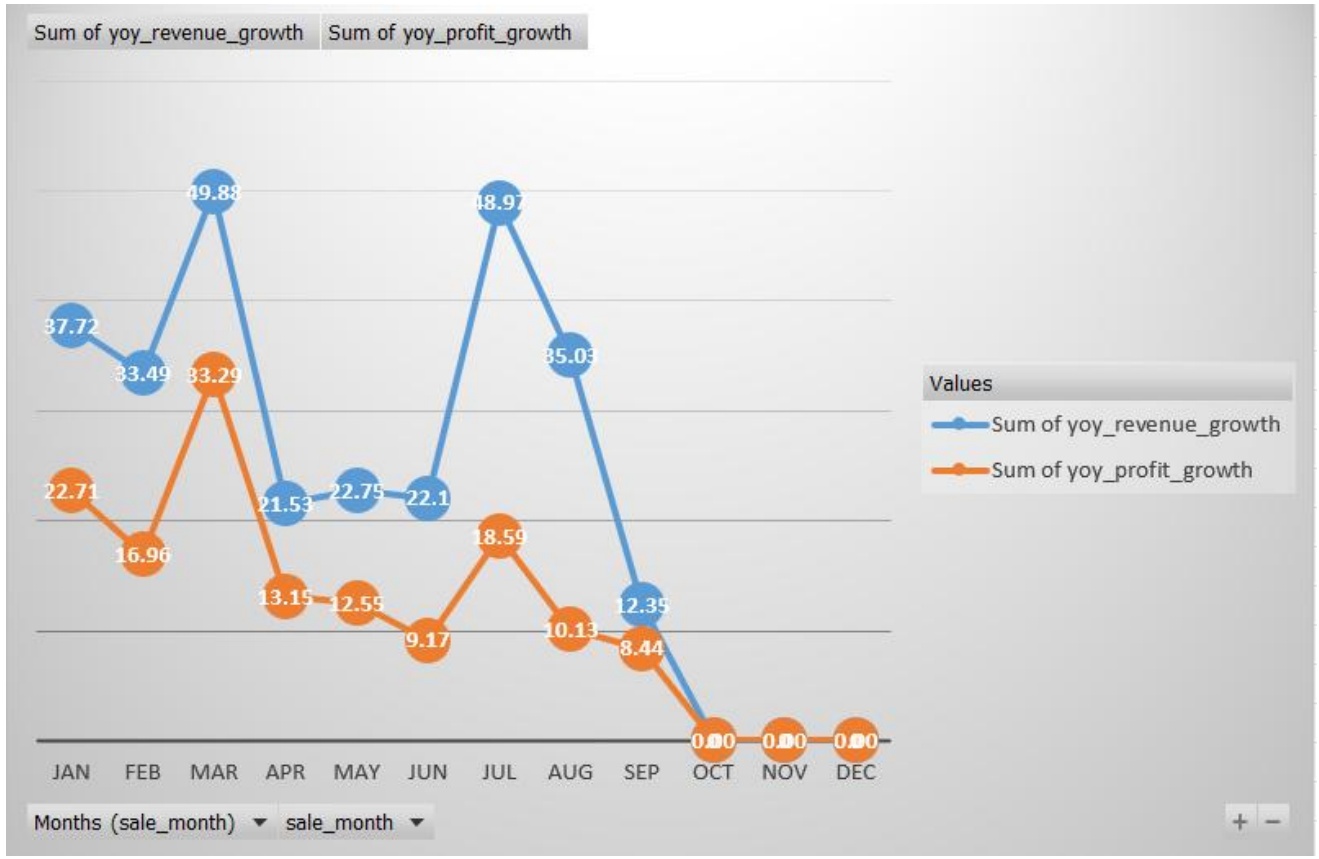
KF 5 – INVENTORY RISK

product_id	product_name	total_units_sold	total_stock	months_of_stock
1	Action Figure	57958	614	0.2
6	Colorbuds	104368	1159	0.2
3	Barrel O' Slime	91663	1282	0.3
15	Hot Wheels 5-Pack	22659	376	0.3
16	Jenga	13054	181	0.3
31	Splash Balls	60248	893	0.3
2	Animal Figures	39089	835	0.4
9	Dino Egg	38236	649	0.4
14	Glass Marbles	37518	795	0.4
18	Lego Bricks	59737	1118	0.4
25	PlayDoh Can	103128	2129	0.4
7	Dart Gun	31588	776	0.5
17	Kids Makeup Kit	24433	606	0.5
21	Mini Ping Pong Set	33604	974	0.6
8	Deck Of Cards	84034	2738	0.7
11	Etch A Sketch	12168	402	0.7
13	Gamer Headphones	17958	637	0.7
19	Magic Sand	60598	1922	0.7
30	Rubik's Cube	45672	1497	0.7

- Products at inventory risk: Action figure and Colorbuds.
- Both are sold the best.
- Therefore it is better to stock up on them before holiday season.

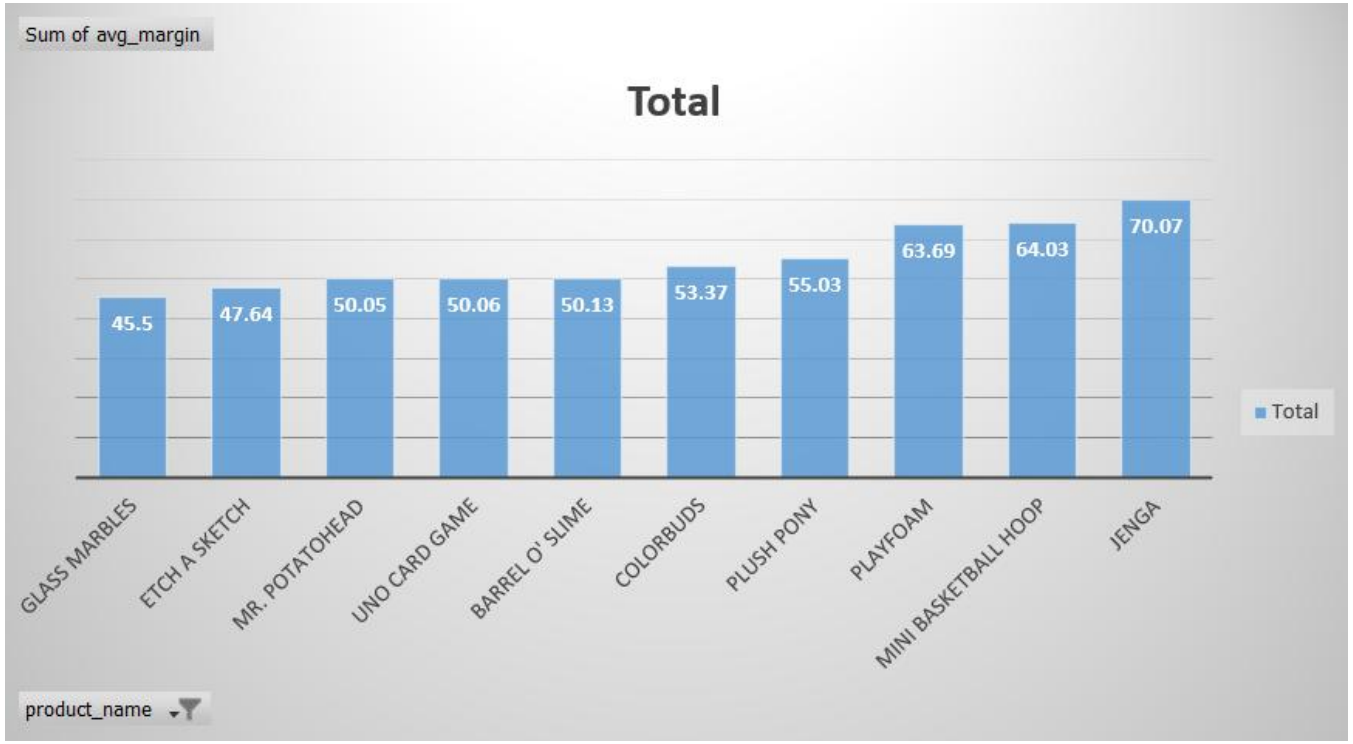
KF 6 – YOY MONTHLY GROWTH

- March and July had the greatest YoY monthly growth.
- New Questions: Why July and March had greatest monthly YoY? Which products where the drivers of this? In which locations those sales wehere made?



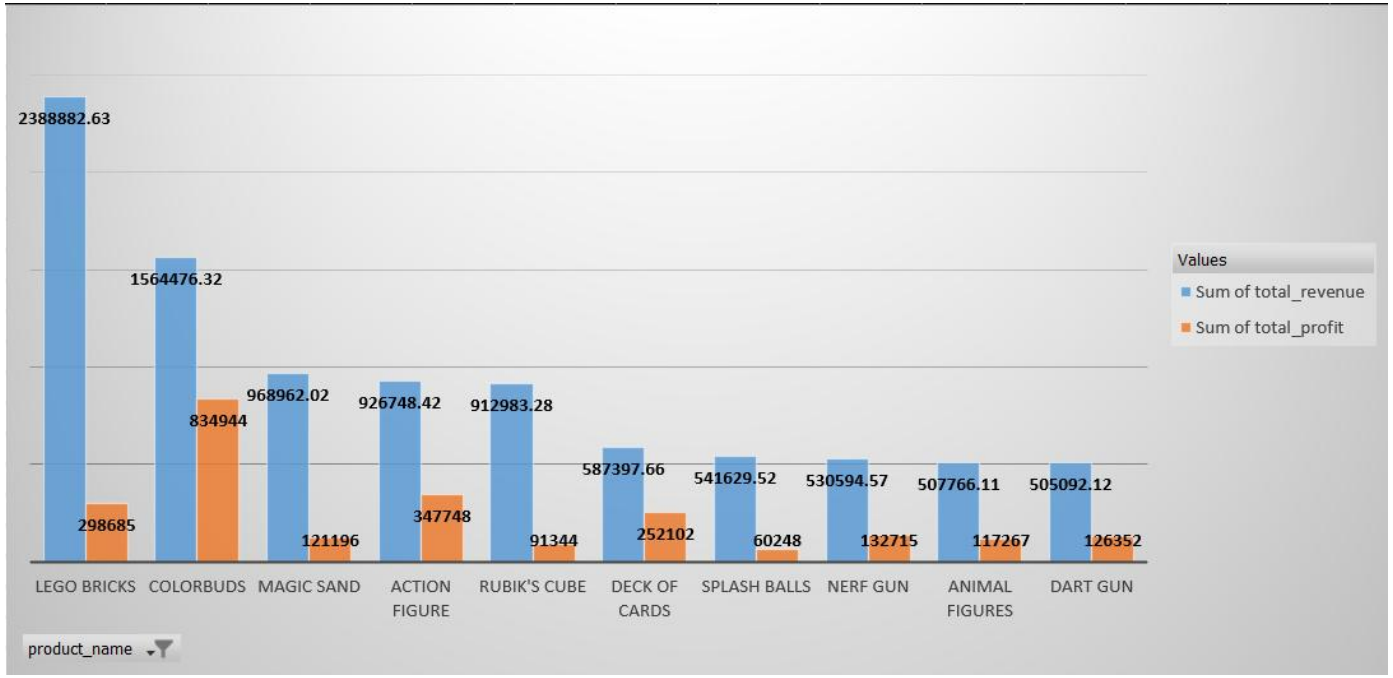
KF 7.1 – PRODUCT PROTFOLIO MATRIX

- JENGA has a 70% profit margin.
- COLORBUDS has 53% profit margin.



KF 7.2 – PRODUCT PROTFOLIO MATRIX

- Lego Bricks, magic sand, rubik's cube has a low profit margin, but generating a lot of profit.
- Seeing this trend we developed a case study.



CASE STUDY – PRICING STRATEGY PROPOSAL

We developed a case study report which is attached to the documents in pdf format.

In this case report a pricing analysis and pricing change and expermentation is proposed.

RECOMMENDED BUSINESS DECISIONS

PRODUCT: Diversify the revenue and profit generation over different products to mitigate the supply chain risks.

PRICING: Experiment with pricing to develope better margins for different products to make greater profit.

SCALING: Get store operating cost data and analyze if it would be profitable to open more stores at Airport locations. Open more shops in Ciudad de Mexico.

INVENTORY: Greatest inventory risks are at December and April, check inventory before those months and stock up if needed. **Stock up on Colorbuds and Action figures.**

FURTHER ANALYSIS POSSIBILIETES

- Analyze marketing performance of Lego bricks and other products.
- Analyze operation cost data to understand how to make more profit.
- Analyze the main competitors in different locations.
- Think about E-commerce capabilities.
- New Questions: Why July and March had greatest monthly YoY? Which products where the drivers of this? In which locations those sales wehere made?

CONCLUSION – WHAT HAVE I LEARNT?

- I have learnt to ingest data with Python.
- I have used Dbt to build staging, intermediate, mart layers and turned inefficient star schema into one big table with pre-calculated metrics.
- Produced a Power BI dashboard, excel report, pptx/pdf report with business decision suggestions.
- Owned all the process from ingestion to deliverables.
- Airflow implementation for pipeline job orchestration.
- Docker implemented.
- Better understanding of data warehousing (Imon to Kimball to OBT).
- How to conduct an end-to-end data pipeline.

- WHAT COULD BE IMPROVED:
  - Airbyte to automate ingestion.
  - New fake sales data created and ingested.

What I would have done differently?

- I would spend more time at first to understand the business case. I would develop a proposal of solutions at first.
- I would find the business questions and business problems first and then develop the technical solutions for it. (Business question -> Do we need a dashboard for it?)
- As now I understand the data team flow. I would better understand the requirements and ask better questions about them.